



CSE623 Machine Learning: Theory and Practice

Topic:

Machine Learning Based Risk Parity Portfolio Construction for Financial Markets

Weekly Report 2

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1. Overview of Project:

Risk parity has emerged as a widely adopted portfolio construction framework that allocates capital by equalizing risk contributions across assets rather than maximizing expected returns. Popularized in institutional asset management as an “all weather” strategy, risk parity portfolios are designed to deliver stable performance across different market regimes. However, the effectiveness of risk parity critically depends on accurate estimation of asset volatilities and correlations.

Recent advances in machine learning and statistical learning theory provide tools for denoising and dynamically estimating covariance structures. Techniques such as shrinkage estimators, random matrix theory based eigenvalue filtering, and hierarchical clustering provides more robust representations of dependency structures (tail risks) for a multi asset portfolio.

This project builds a risk parity portfolio using machine learning techniques. Using daily US financial assets data from 2015–2025, we construct a long-only, fully invested portfolios. We then evaluate their out of sample performance from the year 2020 till 2025 and compare it with a benchmark.

2. Overview of Week 2:

Tasks done till now:

- We initially began with data from Indian equity markets, but later shifted to a broader multi-asset dataset consisting of U.S. equities, U.S. government bonds, commodities, REITs, and currencies. The idea was to move beyond a single-country equity setup and build a more realistic multi-asset framework suitable for implementing a Risk Parity / Equal Risk Contribution strategy.
- All asset return series were merged using an inner join so that only overlapping time periods were retained. This was important to ensure that covariance estimation and portfolio construction were based on consistent data across all assets, without introducing distortions from missing observations.
- To make the analysis dynamic rather than static, the dataset was divided into rolling five-year windows. Within each window, covariance matrices were estimated and portfolio weights were computed accordingly. The portfolio was then rebalanced at the end of each window before moving forward. This creates a structured back testing process that reflects how such a strategy might operate in practice.
- Since Risk Parity relies heavily on the covariance matrix, particularly because risk contributions depend directly on it, so, we estimated covariance matrices using five different methods. These methods form the central comparative component of the study, allowing us to examine how different covariance estimation techniques influence portfolio allocation and overall performance.

3. Work Completed:

- To enhance diversification and align the study with a multi-asset Risk Parity framework, the dataset was expanded from Indian equities to U.S. financial markets. The revised dataset includes S&P 500 equities, currencies, U.S. government bonds, commodities, corporate bonds, and REITs. Data was obtained from Bloomberg Terminal and the “yfinance” library.
- This shift was motivated by the factor-based diversification principle beneath risk parity, which emphasize on equalizing risk contributions across economic risk factors rather than across individual assets. The earlier Indian equity dataset limited economic factor exposure. Additionally, U.S. markets have greater liquidity and depth, making them more suitable for our analysis.
- Daily data post-2015 period (10 years) was collected, resulting in 474 constituents across asset classes such stocks, bonds, commodities, currencies and REITs. Observations prior to 2015 were excluded, and securities with missing values were removed. The same filtering process was applied to the Bloomberg dataset. An inner join was performed to retain only securities with complete overlapping observations.
- To build a dynamic portfolio allocation model, the cleaned dataset was divided using a rolling window covariance estimation to derive weights. Portfolio weights will be computed within each window and will be evaluated over a subsequent one month holding period.
- For covariance matrix estimation, a five-year rolling historical window was employed to maintain certain degree of recency bias and not expanding window method.
- To ensure robustness and address estimation error, five covariance estimation techniques were implemented:
 - (i) the historical sample covariance estimator,
 - (ii) the Ledoit–Wolf shrinkage estimator,
 - (iii) Oracle Approximating Shrinkage (OAS),
 - (iv) constant shrinkage covariance estimation, and

(v) targeted shrinkage covariance estimation.

- We referred to Dany Cajas's book *Advanced Portfolio Optimization* to understand and implement covariance estimation methods.
- We referred to Thierry Roncalli's paper *Risk Parity Portfolios with Risk Factors* to understand the purpose of risk parity and accordingly went ahead with including asset classes which increase our exposure to multiple economic risk factors that can be potentially orthogonal to each other.

4. Challenges encountered:

- Finding the right dataset that holds up to the philosophy of Risk Parity, which is to invest money in different sources of economic risk factors and attain diversification that is safe during the times of negative movements in the market. Also, liquidity of the market was an important factor for choosing US markets and not choosing the Indian market.
- Finding the right methods for estimating covariance matrices from the literature.

5. Plan for the Week 3:

- In the next phase of the project, implicit risk factors will be extracted using Principal Component Analysis (PCA) applied to the asset return matrix. PCA will facilitate dimensionality reduction while capturing dominant sources of systematic variation across asset classes.
- The resulting principal components will be interpreted as latent risk factors, and weights will be assigned based on their explained variance and associated risk contribution.
- These factor weights will subsequently be translated into asset-level allocations to construct the portfolio with optimal weight.

References:

1. Cajas, D., *Advanced Portfolio Optimization*: Springer (2025)
2. R. Roncalli and B. Weisang, "Risk Parity Portfolios with Risk Factors," MPRA Paper.